WESTERN SYDNEY UNIVERSITY

International College

Object Oriented Analysis Project Report

Beach Holiday Park System (BHPS)

2024.3

Submitted By

Group Name: WXYZ.

Student No.	Name	Package
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Executive Summary

Beach Holiday Park (BHP) is a fast-paced growing resort where families, friends, work groups and every group can find their comfort and coziness. Due to the numerous amounts of people, the park has been considering expanding the terrain they control. With this, the shareholders have brought about the idea of transforming into a complete scaling e-business, where not only bookings are made via internet, but also the entire functionality of the company. In other words, staff members including all the managers, receptionists, contractors, restaurants staff, suppliers, and even customers will communicate in an optimum and time saving manner. For instance, phone code connections, instant reparation alerts, automatic system synchronization.

Introduction

This report includes diagrams and documentations for clear understanding of getting the system online. It includes Actor List & Documentation, Use-Case Documentations, Use-case Diagrams, Activity Diagrams, Class Documentations, Class Diagrams, Sequence Diagrams, Interaction Overview Diagrams, State-machine Diagrams. This also includes meeting minutes of the members who participated into it and the time consumed in preparing all the diagrams and documentations.

Scope:

This report includes deliverables completed for given case study on Beach Holiday Park (BHP). This file includes work of Iteration 1 & Iteration 2.

Project Plan & Minutes:

Minutes of the Project Team Meeting of 11/01/2024

Meeting No: 2 Held at: _____ Zoom Class_____

Chairperson: _____ Jung, Williams_____

Team members in attendance: Jia, Yurong – Jung, Williams

Minutes taken: **1 hour 15 minutes**

Agenda items discussed:

Iteration 2

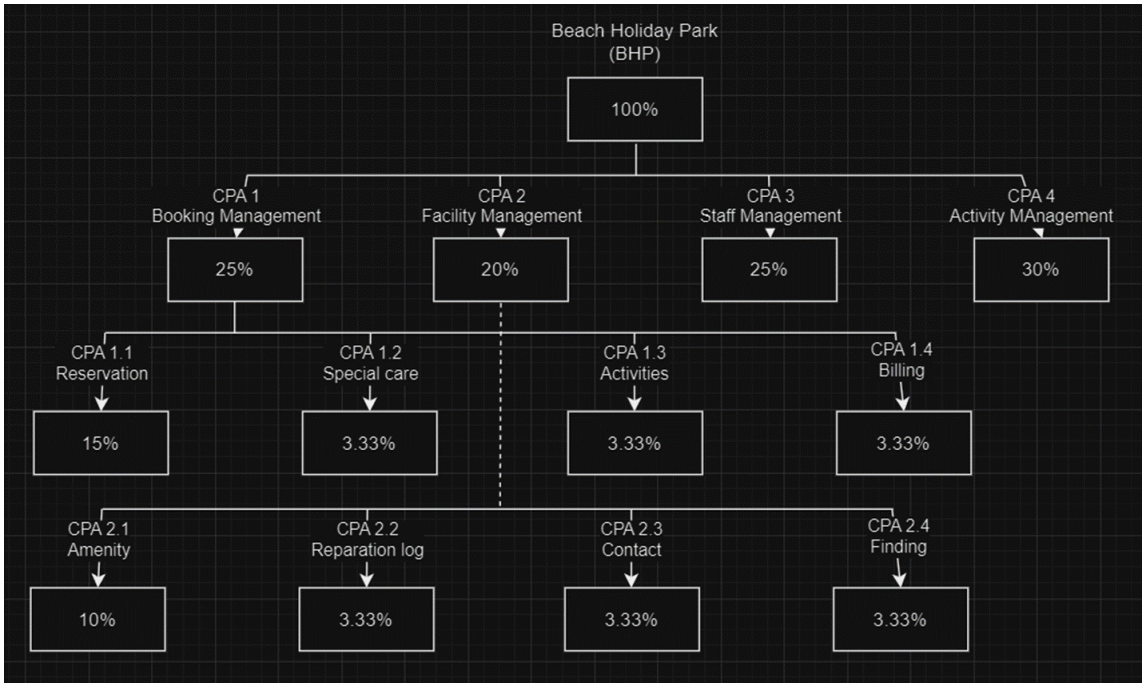
Action Plan:

Item	Description	Who	When
1	Discussing project meeting minutes	group	
2	Submission of iteration 2	group	21/01/2024

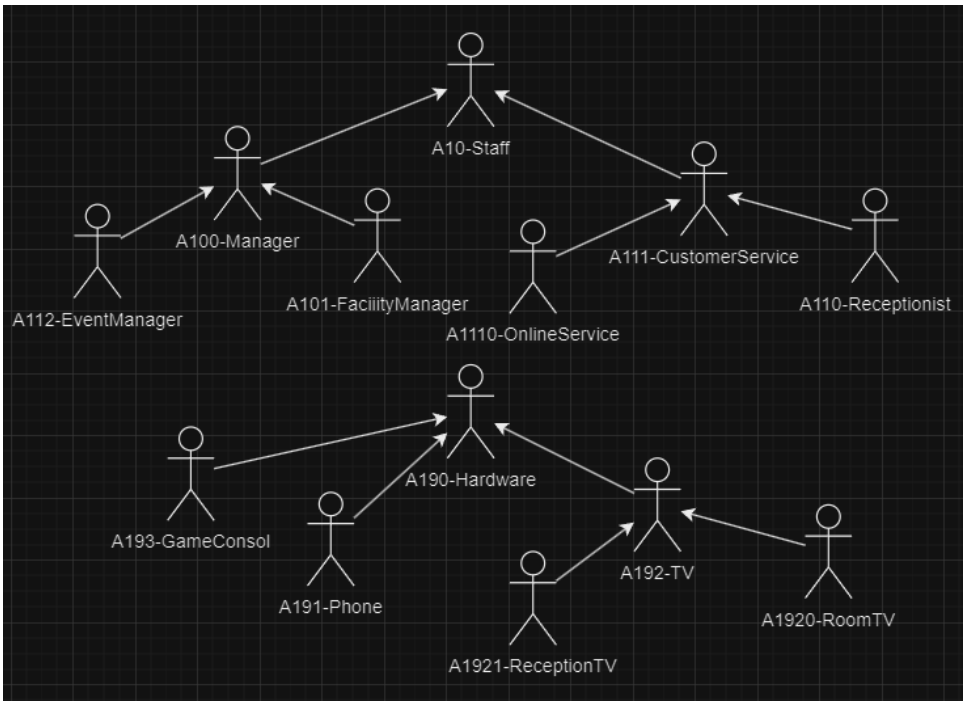
Critical Requirement Analysis:

Business Objective: Elevate the system functionalities to satisfy BHP's desired specifications.

- Ensure the creation of a cost-effective, scalable and malleable to future updates software system. (What?)
- By connecting the entirety of the park in an auto-synchronize system. (How?)
- To maximize customer satisfaction and easy-usability and minimize avoidable work to engage an optimum time management. (Why?)



Actor List Diagram



Actor Documentation

Booking management Package (Total 2)

Actor Thumbnail:	**Actor: A112-BookingManager**
Actor Type & Stereotype:	Responsible for managing and maintaining the booking process in the system.
Actor Description:	The objective is to ensure an efficient booking process. This includes confirming, canceling and changing reservations, handling special guest preferences, and generating reports related to booking trends, revenue and occupancy.

Actor Relationships:	A02-Mamber A10-Staff A120-SystemAdministrator A101-FacilityManagement
Interface Specifications:	UI010-ConfirmReservation UI020-GenerateBookingReports
Author & History:	Yurong Jia
Reference Material:	Australia Government rules

Facility Management Package (Total 2)

Actor Thumbnail:	**Actor: A101-FacilityManager**
Actor Type & Stereotype:	This actor represents the person who oversees registering and updating the designated facility.
Actor Description:	Its objective is to ensure optimum customer experience. This includes facility tidiness, restaurant and entertainment effective functionality, as well as problem solving and correct following of regulations and standards.
Actor Relationships:	•Several actors surfacing from this one: 1.A006-ReparationContractor 2.A060-ReparationLog 3.A10-Staff •The actor will interface with the following use cases: 1.UC600-ReportFailure 2.UC060-ReparationDetail
Interface Specifications:	•UI0600-ReportProblem •UI0060-ReparationLog
Author & History:	Williams Jung

Reference Material:	Australian Standards, legislative and work health, and safety requirements

Actor Thumbnail:	**Actor: A191-Phone**
Actor Type & Stereotype:	This actor represents the telephone lines which are mandatorily located in every single room of every single building.
Actor Description:	Its objective is to ensure connectivity between not only room to room, but room to receptionist, restaurants, attractions, and much more. These are introduced with specific codes which facilitate and speed up customers communicating with the external room personas.
Actor Relationships:	•Several actors surfacing from this one: 1.A101-FacilityManagement 2.A110-Receptionist 3.A10-Staff •The actor will interface with the following use cases: 1.UC600-ReportFailure 2.UC121-OrderFood
Interface Specifications:	•UI0600-ReportProblem •UI0120-RoomServiceMenu
Author & History:	Williams Jung
Reference Material:	Australian Standards, legislative and work health, and safety requirements

Use Case Documentation

Booking management Package (Total 4)

1)

Use Case Thumbnail:	**UC800-FacilityAvailable**
Package:	Booking Management
Actors:	A02-Member A10-Staff A120-SystemAdminstrator
Use Case Description:	Members should be able to check the real-time availability of facilities, including current and future time slots.
Pre-Condition:	Members regist.
Post-Condition:	Members search for specific facilities and activities.

Basic Flow:	1. Members register their account. 2. Members check the real-time availability of facilities, including current and future time slots. 3. Members search for facilities and activities. 4. Members make a reservation for accommodations, facilities, and activities. 5. An email is sent to the member to provide detailed information about it.
Alternate Flow:	A1. If the facility is unavailable, members will be prompted to select an alternative time A2. If the facility is available, the system provides the option of member confirmation and enters the payment interface.
User Interface Specifications:	UI090-SearchFacility UI010-ConfirmReseration
Author and History	Yurong Jia

2)

Use Case Thumbnail:	**UC700- ManageReservation**
Package:	Booking Management
Actors:	A02-Member, A10-stuff, A0301-Resturant, A031-RoomNumber, A0311-VIPRoom

Use Case Description:	A02-Member manage their reservation including confirming the reservation by paying for the accommodation, cancelling, or changing the reservation. A10-Stuff handle it on behalf of the members.
Pre-Condition:	Members should complete their payment.
Post-Condition:	Members want to change their room.
Basic Flow:	1. The potential member logs into the system and selects the "View Reservations" option. 2. The system displays a list of pending reservations for members to review. 3. The member selects a reservation that needs confirmation and clicks the "Confirm" button. 4. The booking manager receives notification of the confirmed reservation. 5. The system sends a confirmation email to the member and updates the booking manager's dashboard. 6. The reservation is now marked as "Confirmed," and the member receives a confirmation email with details. 7. Generate booking results and booking trends and send them to booking manager.
Alternate Flow:	A1- After confirming a reservation, the member decides to cancel. 1. The member selects the "Cancel Reservation" option in their dashboard. 2. The system prompts the member to provide a reason for cancellation. 3. The system updates the reservation status to "Canceled" and adjusts the availability status. 4. The member receives confirmation of the cancellation. A2- The potential member tries to confirm a reservation, but there is insufficient availability. 5. The system displays an error message informing the potential member about the unavailability. 6. The potential member is prompted to select an alternative date or accommodation type. 7. The system updates the availability status accordingly and allows the potential member to choose an alternative.

User Interface Specifications:	UI010-Confirm Reservation UI020-GenerateBookingResults UI606-ReportFailure UI707-Booking Cancellation
Author and History	Yurong Jia

3)

Use Case Thumbnail:	UC80-Payment
Package:	Booking Management
Actors:	A10-customer, A20-Staff
Use Case Description:	This use case enables the customer to pay for their apartment using various payment methods.
Pre-Condition:	Customer must select the one of the options provided.
Post-Condition:	Payment must be available.

Basic Flow:	1. Members go to the website to confirm reservations and make a payment. 2. Three payments options are provided from which the customer selects one to make payment. (A1) 3. The system asks members to provide payment details to make payment. 6. Members provide necessary information to make payment. (A2) 7. System asks member to confirm payment. 8. Members make payment by confirming. 9. System confirms the payment was made and sends an email with the details of the flight.
Alternate Flow:	A1. If the customer cannot use any of the payment methods, he/she is asked to email a staff member to consult about the payment methods. A2. If the information provided by the customer is wrong, the system asks to provide them again.
User Interface Specifications:	UI011-PaymentForm
Author and History	Yurong Jia

Facility Management Package (Total 4)

Use Case Thumbnail:	**UC600-Reparation**
Package:	Facility
Actors:	A101-FacilityManager, A01-Customer, A006-ReparationContractor
Use Case Description:	This use case takes care of needed reparations and reports of reparations. A A101-FacilityManager is assigned to have control over every facility from Beach Holiday Park (BHP). Since the immensity of the park, the territory is separated and named like animals.
Pre-Condition:	The need for a reparation in any facility and any object
Post-Condition:	Reparations are performed, reported and status updated.
Basic Flow:	1. Customer or staff reports something to be fixed 2. Facility manager checks and follows procedure (A1) 3. Reporting area name, building number, room number, date, and what needs to be repaired 4. A specialized contractor is called, and if supply is to be acquired, do so too. 5. Status is then updated with necessary information like date, who fixed it, and expenditure produced (A2)
Alternate Flow:	A1- Following procedure may be assigned by the company or decided by the facility manager. For instance, if it happens frequently, a different trajectory on how to solve the problem might be taken upon. A2- The park already employs contractors; however, if an object needs to be replaced due to malfunction or because it became obsolete, another should be acquired. The invoice for this fee is then attached to the report, for the company to take care of.
User Interface Specifications:	UI606-ReportFailure
Author and History	Williams Jung

Use Case Thumbnail:	**UC999-UpdateFacility**
Package:	Facility
Actors:	A101-FacilityManager
Use Case Description:	This use case documentation is operated and updated by A101-FacilityManager. This is to avoid data redundancy, and to target similarity in information for it to be understood by everyone.
Pre-Condition:	Necessity to update or remove a facility report.
Post-Condition:	Facility report has been updated.
Basic Flow:	1. A101-FacilityManager enters the facility log document. 2. System displays all the facilities. 3. The manager then proceeds with updating, adding or removing certain facilities. (A1) 4. Reports are saved and automatically updated in every document managed by each manager or area. (A2)
Alternate Flow:	A1- Since this is a rapidly scaling Park, it might be necessary to have several documents which include specific facilities like restaurants, attractions, VIPRooms, EconomicRooms, etc. A2- This also needs to be notified by sending individual messages to every manager and/or staff member. To minimize errors and confusion between workers.
User Interface Specifications:	UI989-FacilityReport
Author and History	Williams Jung

Use Case Thumbnail:	**UC900-SupplyManagement**
Package:	Facility
Actors:	A100-Manager
Use Case Description:	This use case documentation is operated and updated by each respective manager, like facility, restaurant, cleaning, and more. These are all the suppliers which are used by all the managers in the Park. This must be easy to modify and must be used by all the

	managers, where each will either dispose of their own log or have one report with divisible information.
Pre-Condition:	Necessity to update or remove suppliers and supplies.
Post-Condition:	The supplier report has been updated.
Basic Flow:	5. The manager enters the suppliers and supply log document. 6. The system displays all the suppliers and supplies available divided into each respective category. (A1) 7. The manager then proceeds with updating or removing suppliers or supplies. 8. Reports are saved and automatically updated in every document managed by each manager or area. (A2)
Alternate Flow:	A1- To avoid confusion and errors, all the inventory including the suppliers are to be separated into each specific category. This also allows a better scaling procedure and data integrity.
User Interface Specifications:	UI901-SuppliersReport
Author and History	Williams Jung

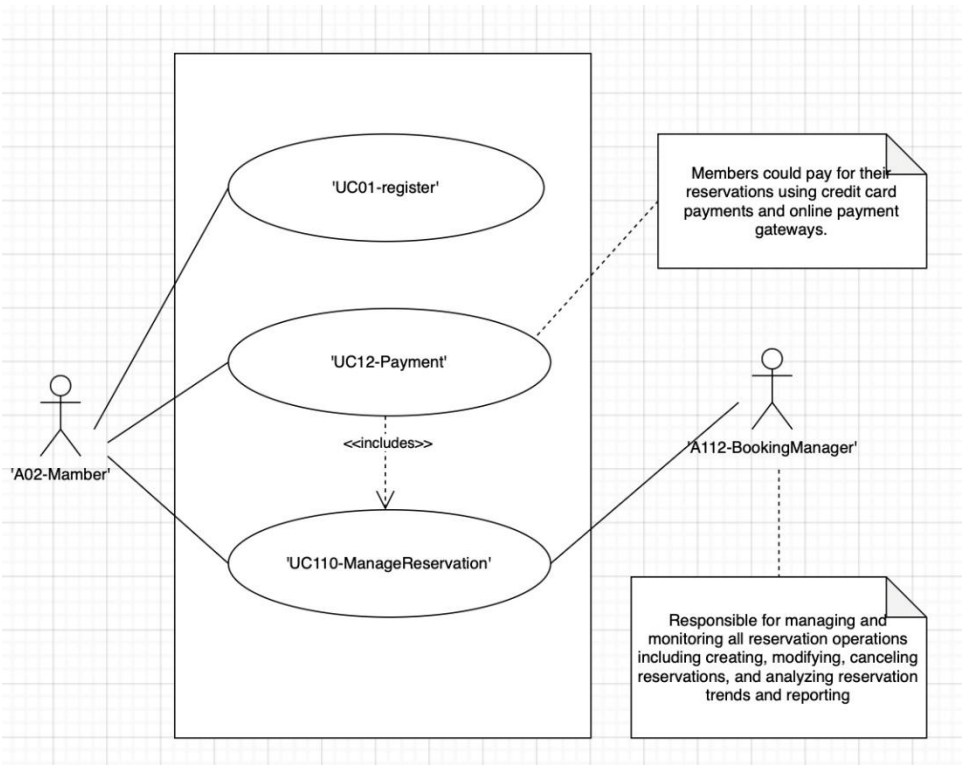
Use Case Thumbnail:	**UC9090-SupplyOrdering**
Package:	Facility
Actors:	A100-Manager
Use Case Description:	This use case documentation is operated and updated by each respective manager, like facility, restaurant, cleaning, and more. This refers to the process of supply ordering in each area, like alimentary prodcuts for the restaurant managers, cleaning materials for cleaning managers, and more.
Pre-Condition:	Necessity to stock up supplies.
Post-Condition:	Order has been confirmed and invoice received.
Basic Flow:	1. The manager checks stock availability. 2. The manager makes a list of supplies that need to be ordered. (A1)

	3. The manager then sends an order to each supplier and receives an order confirmation. 4. The suppliers send the invoice, and the managers check for errors. (A2) 5. The invoice is signed by the manager and sent to finance.
Alternate Flow:	A1- This needs to be done by only one person per area to avoid over-ordering, wasting more money than the budget, and not using stock room wisely. A2- The same manager that sent the orders must check the invoice, and then compare it when items arrive.
User Interface Specifications:	UI9090-SupplyOrdering
Author and History	Williams Jung

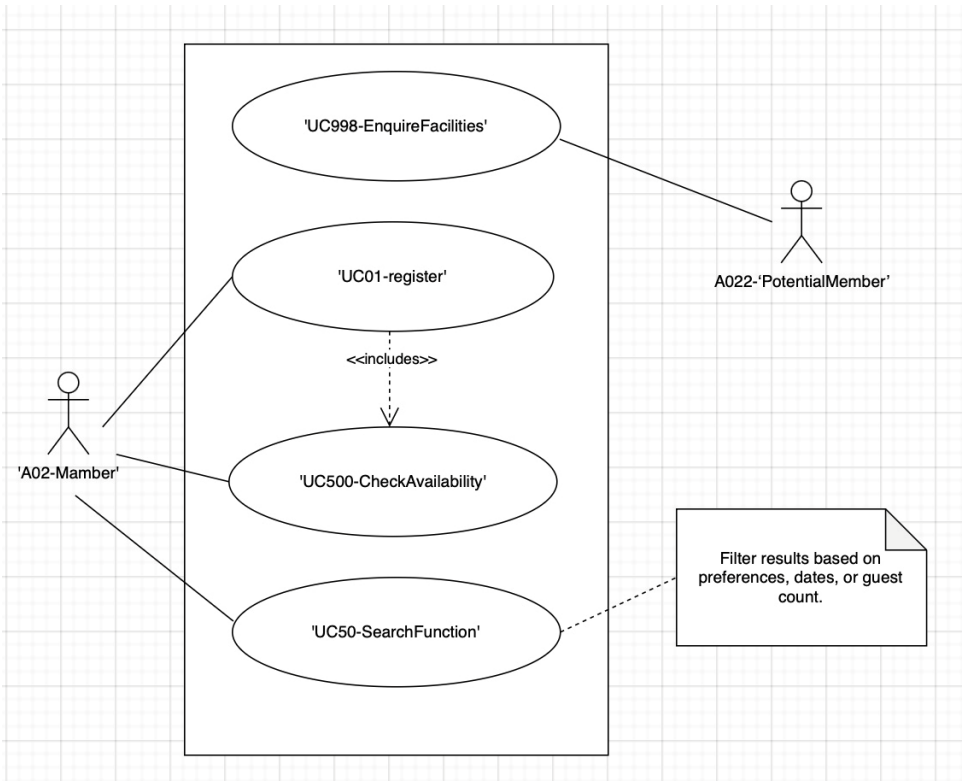
Use Case Diagram

Booking management Package (Total 2)

Manage Reservation

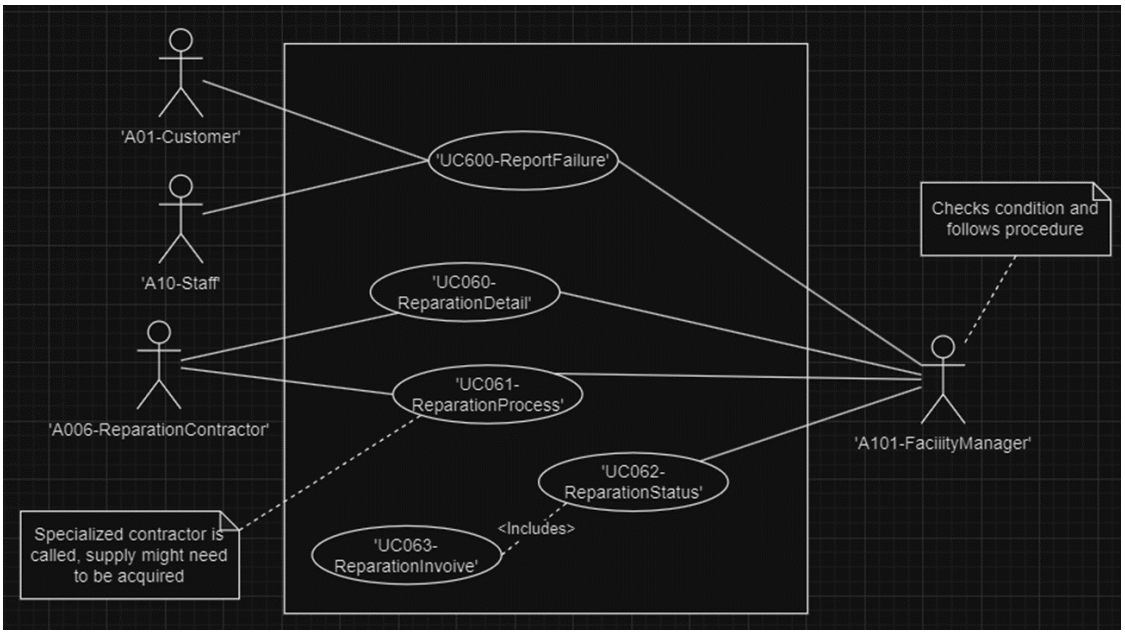


Check facility availability

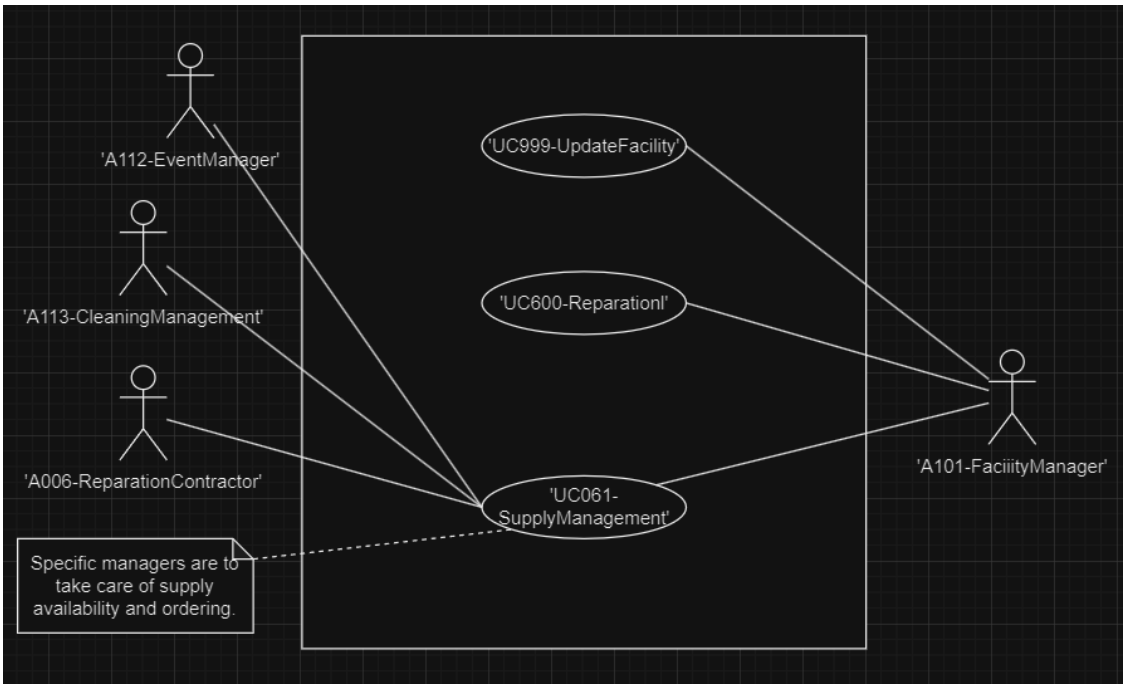

'UC998-EnquireFacilities'
'UC01-register'
<<includes>>
'UC500-CheckAvailability'
'UC50-SearchFunction'
'A02-Mamber'
A022-'PotentialMember'
Filter results based on preferences, dates, or guest count.

Facility Management Package (Total 2)

Reparation process



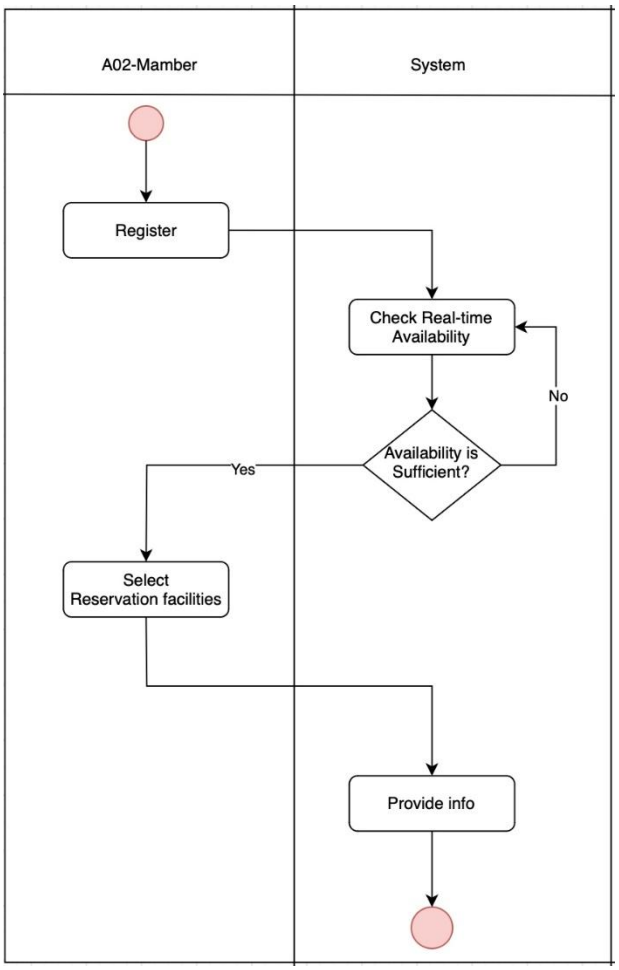
Updating or removing facilities



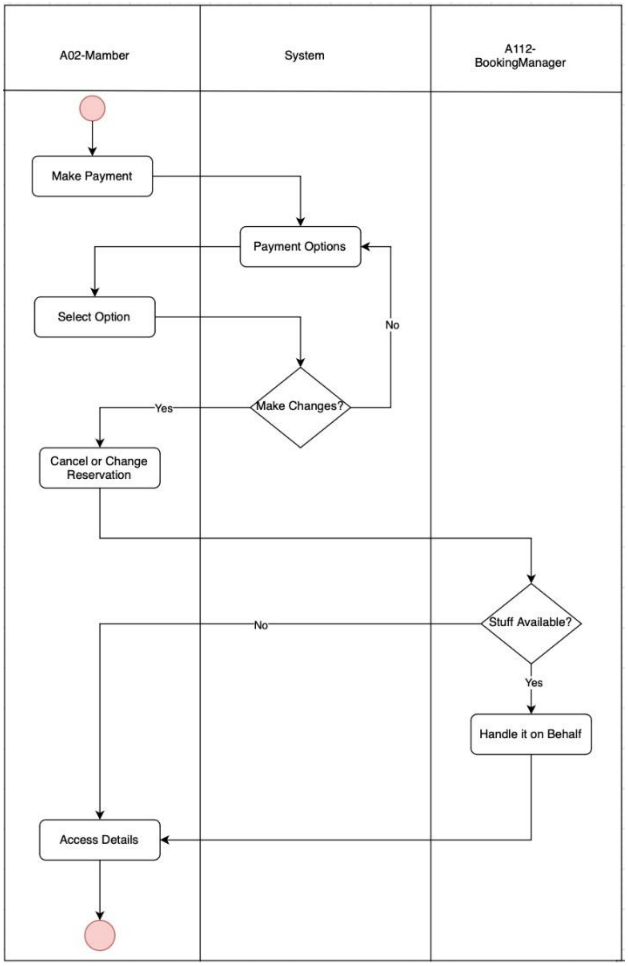
Activity Diagram

Booking management Package (Total 4)

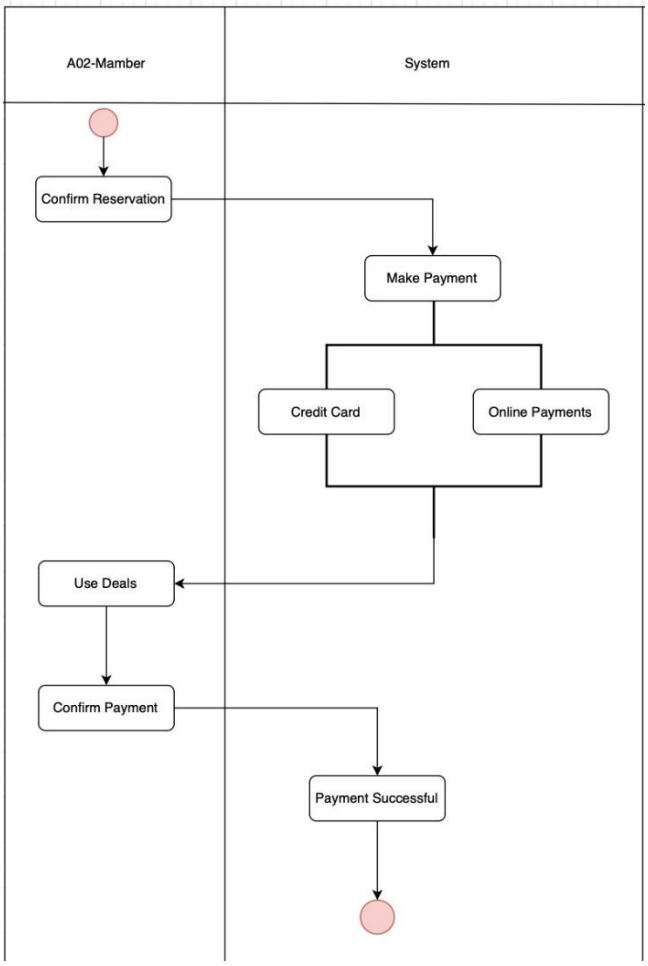
Enquire facilities



Manage Reservation

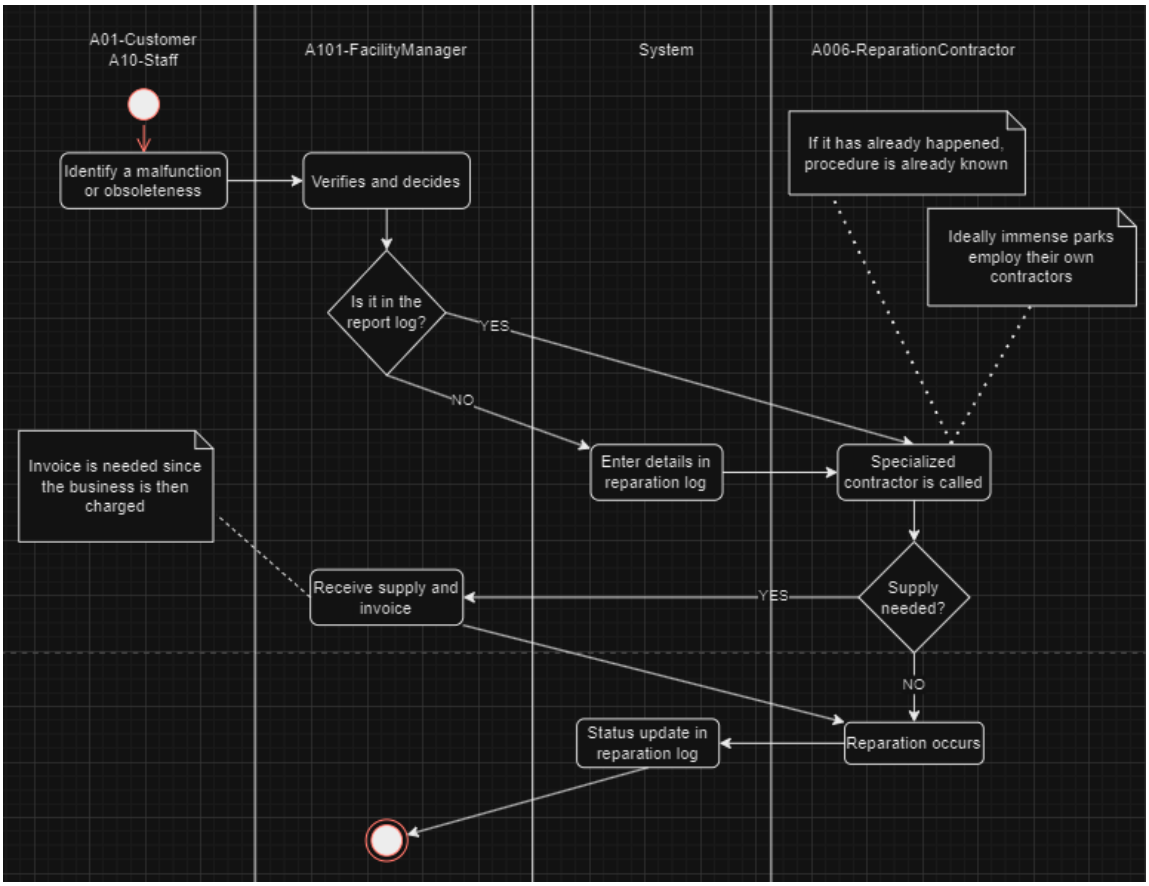


Payment

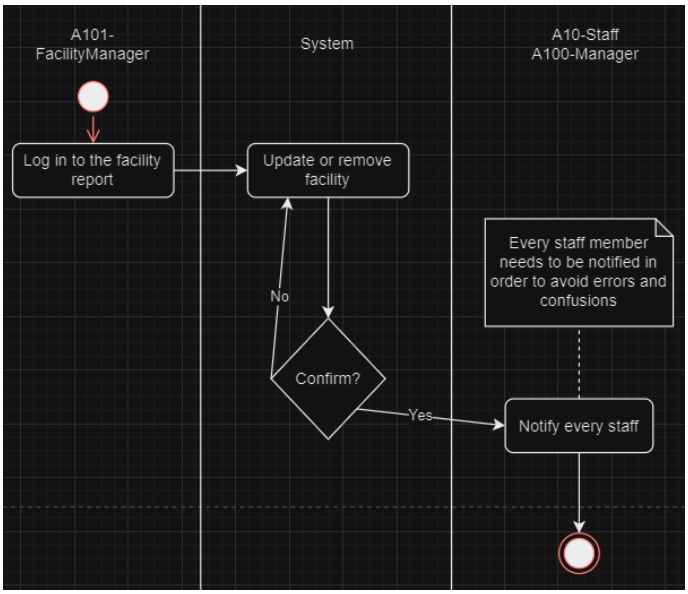

A02-Mamber
System
Confirm Reservation
Make Payment
Credit Card
Online Payments
Use Deals
Confirm Payment
Payment Successful

Facility Management Package (Total 4)

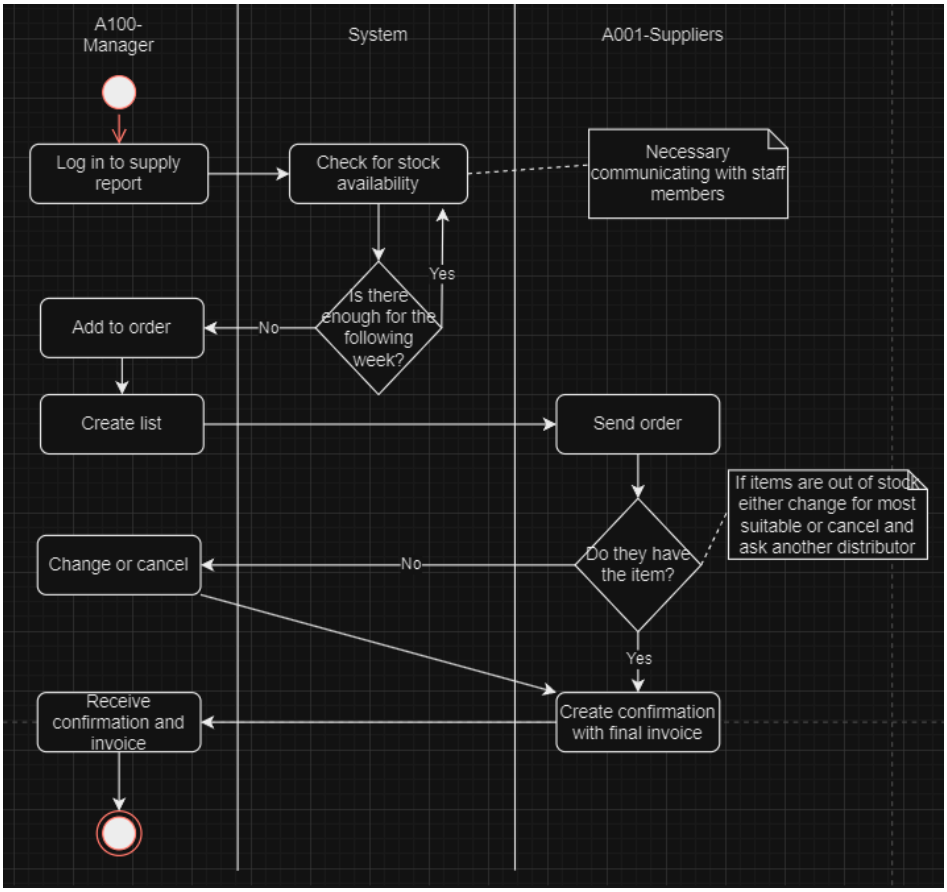
Reparation Report



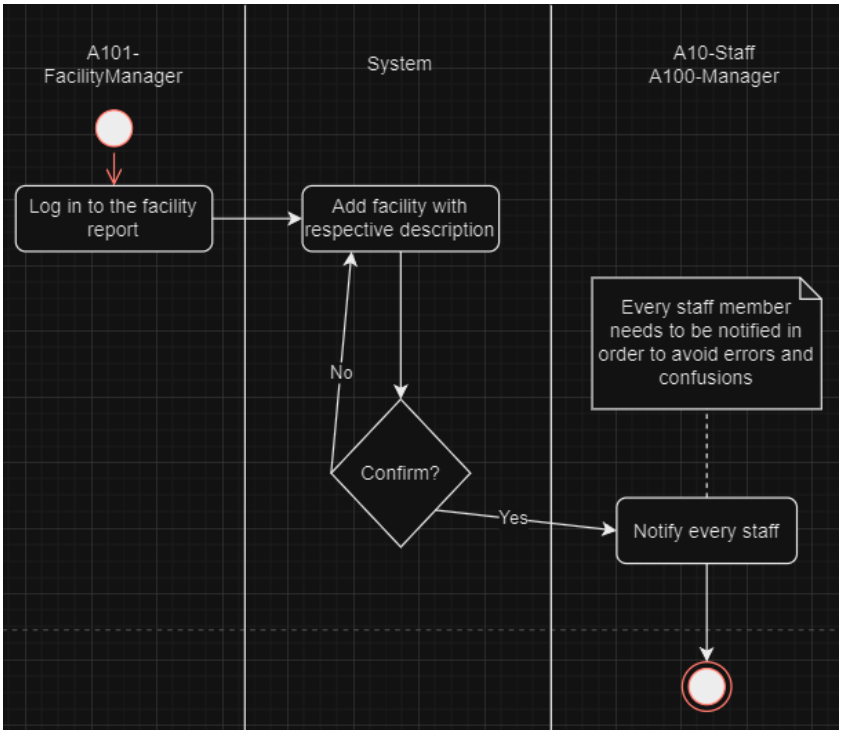
Updating facilities



Supply ordering

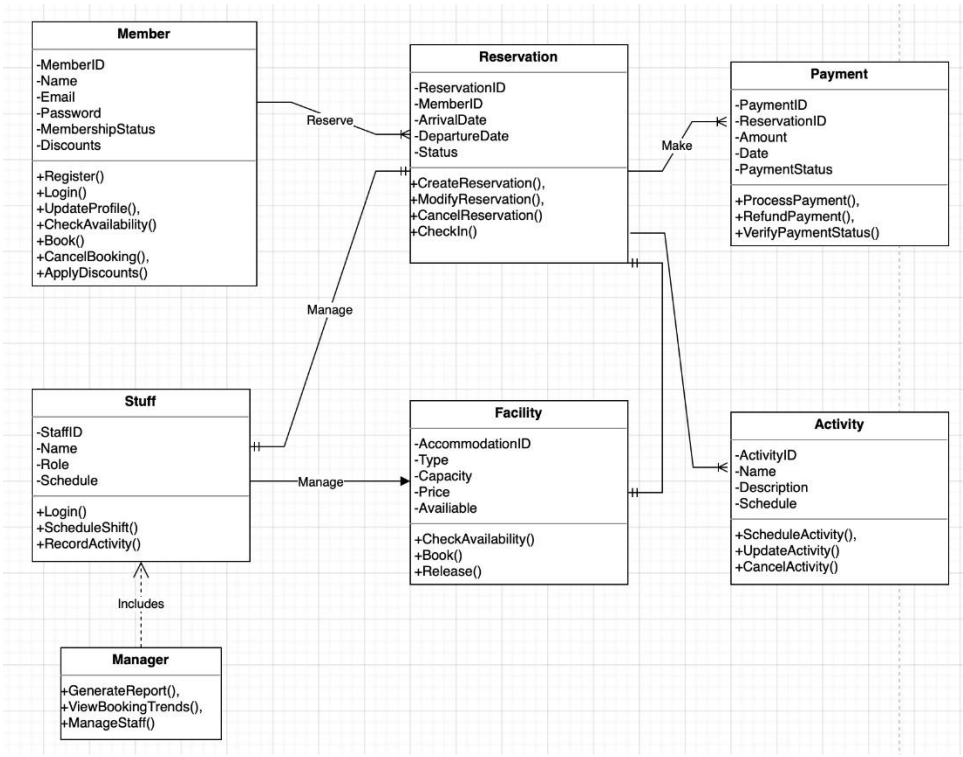


Adding facility

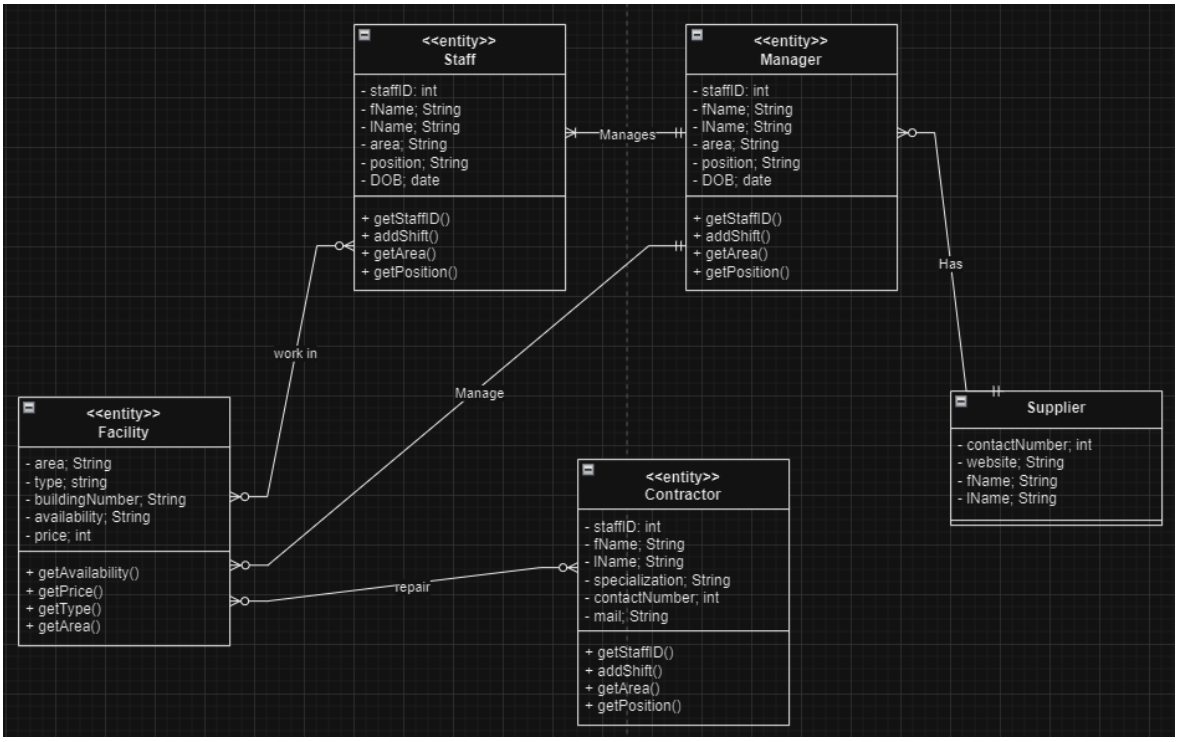


Class Diagram

Booking management Package (Total 1)

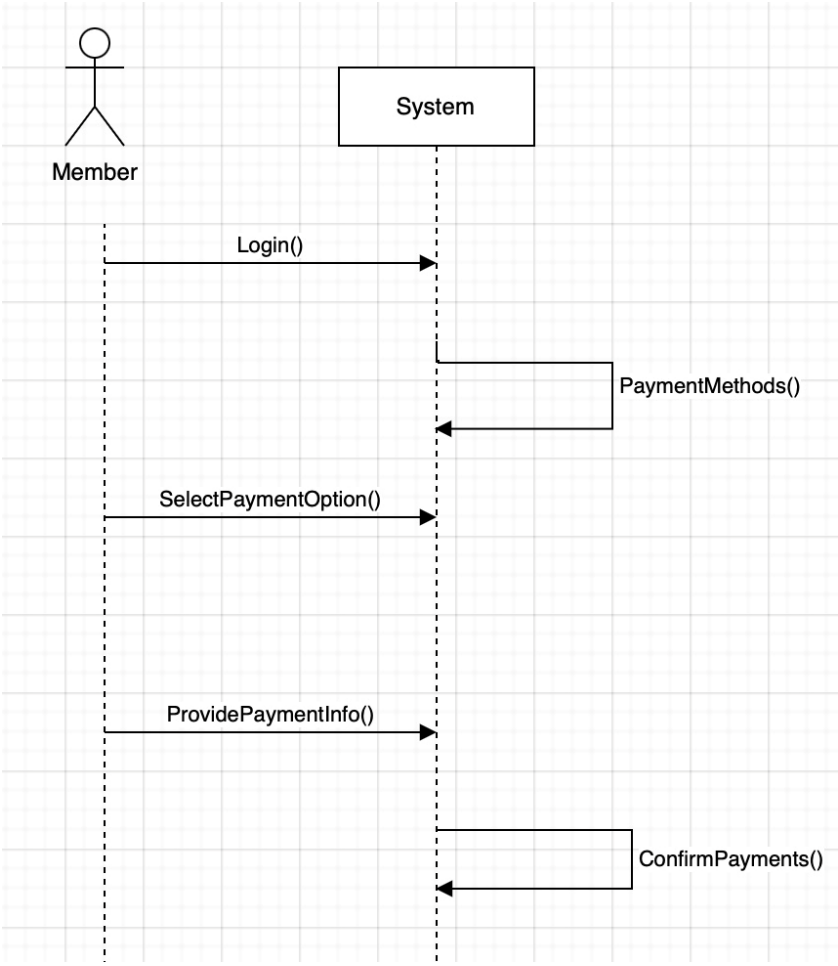


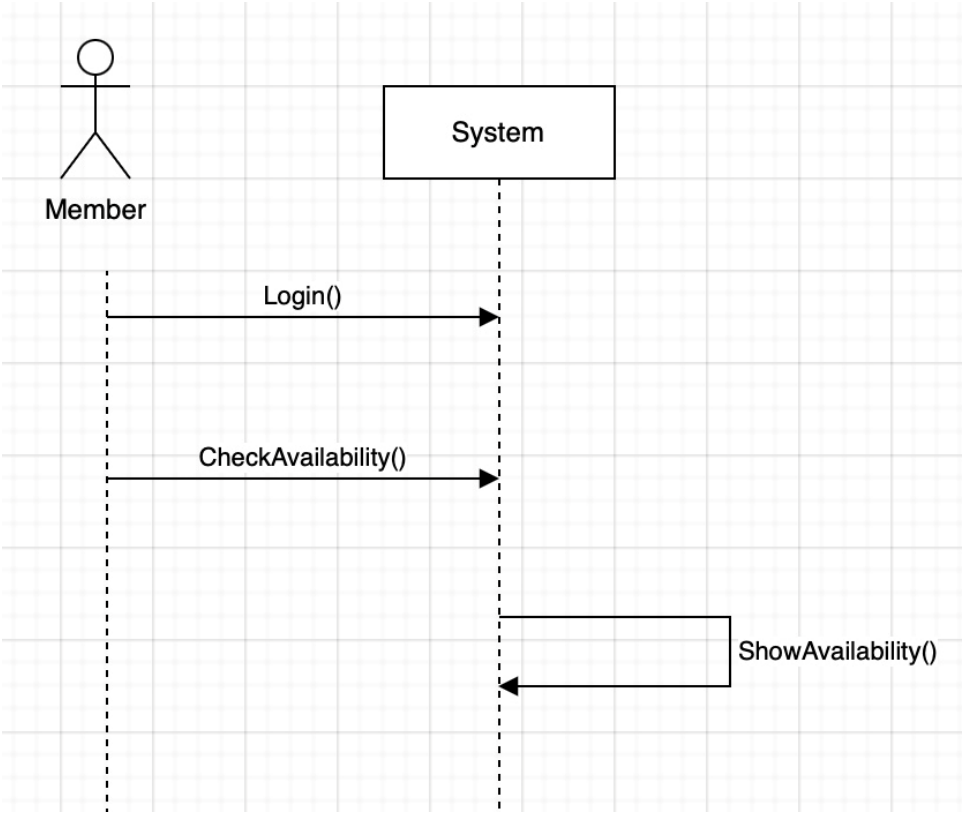
Facility Management Package (Total 1)



Sequence Diagram

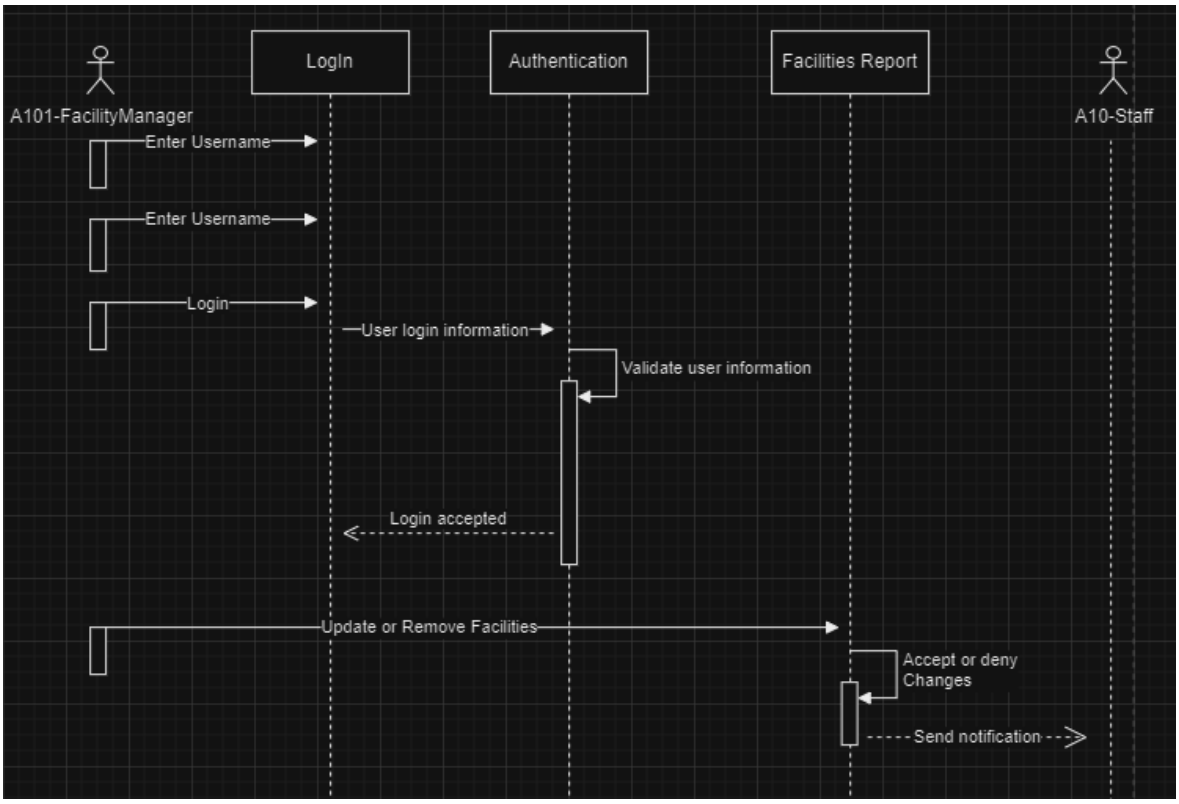
Booking management Package (Total 2)


Member
System
Login()
PaymentMethods()
SelectPaymentOption()
ProvidePaymentInfo()
ConfirmPayments()

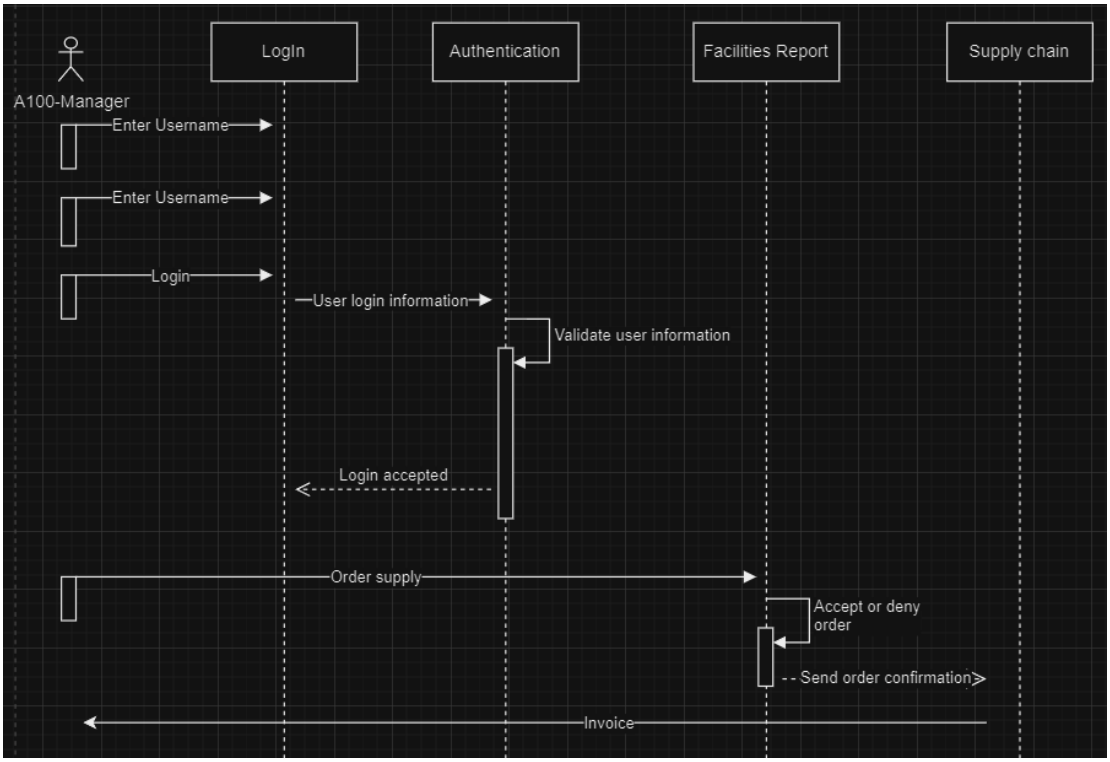


Facility Management Package (Total 2)

Updating or removing facilities

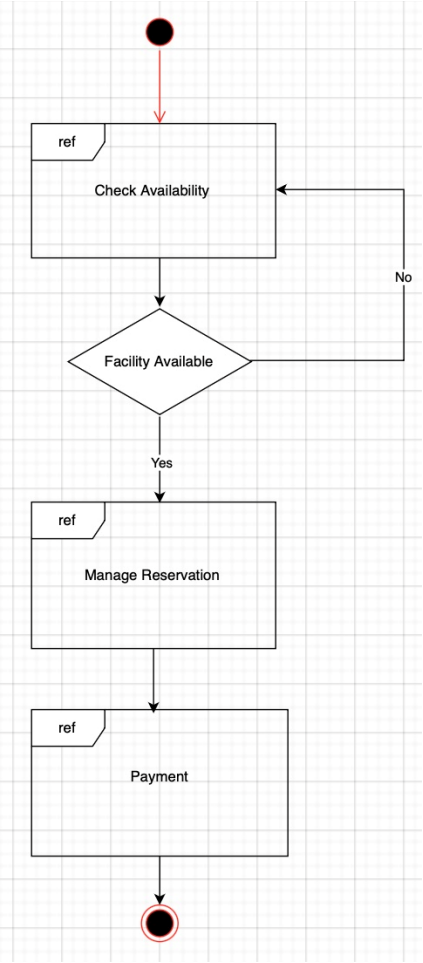


Ordering supply



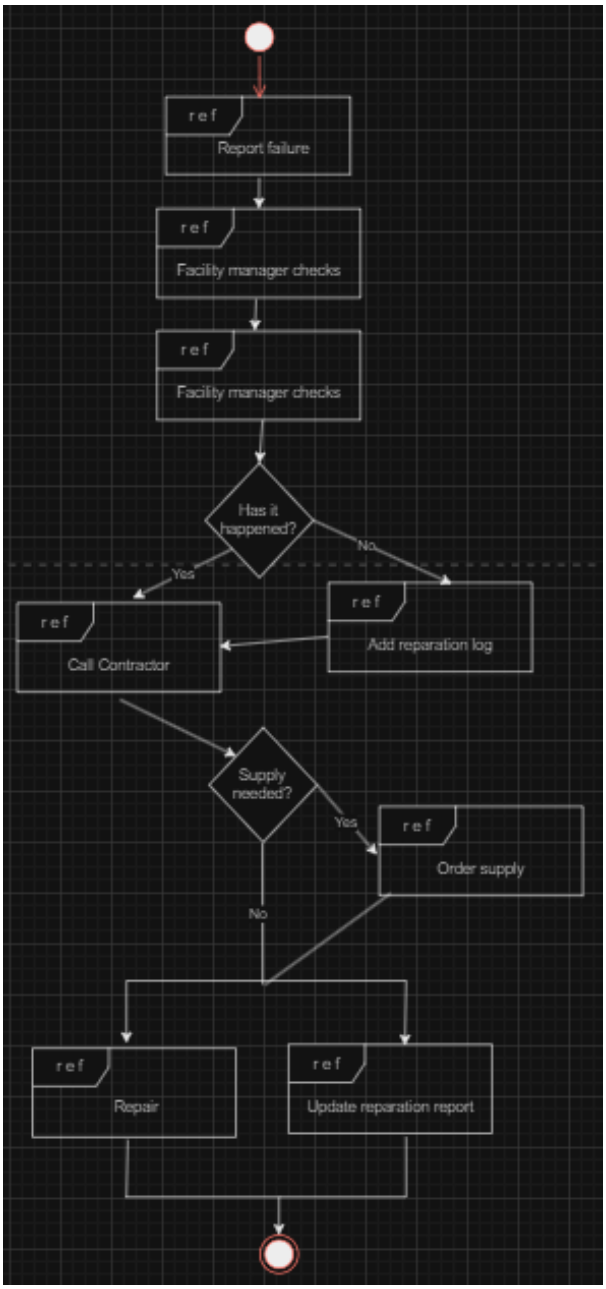
Interaction Overview Diagram

Booking management Package (Total 1)



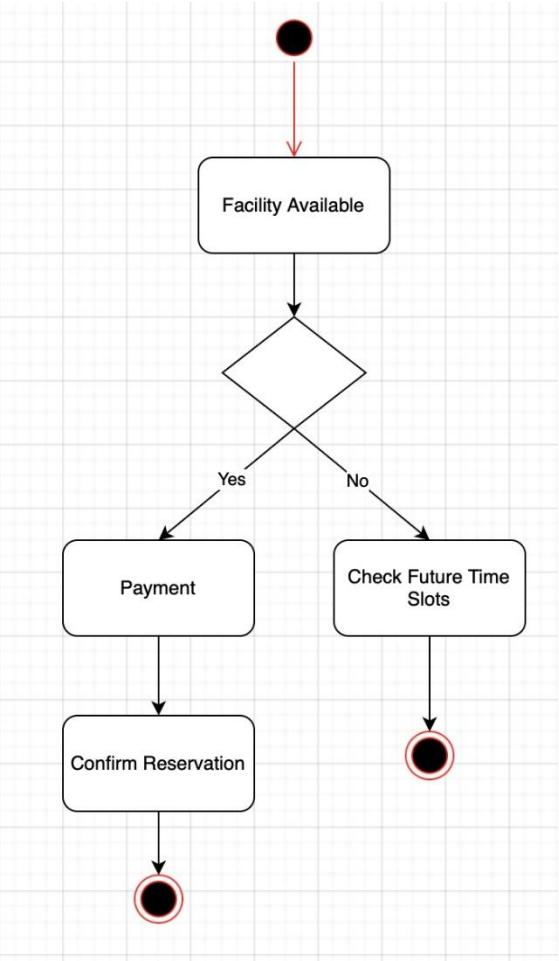
Facility Management Package (Total 1)

Repair

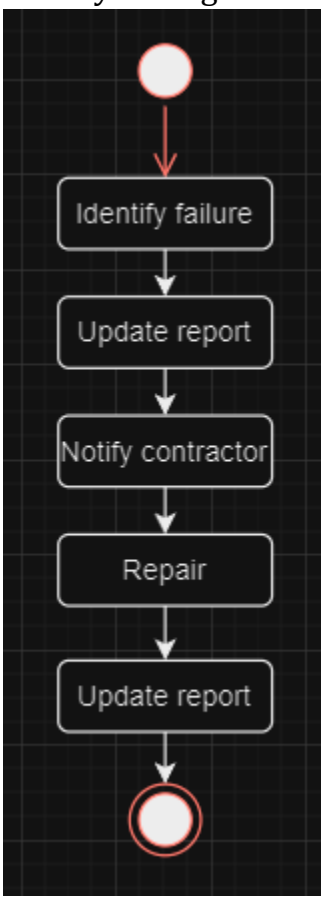


State Machine Diagrams

Booking management Package (Total 1)


Facility Available
Yes
No
Payment
Check Future Time Slots
Confirm Reservation

Facility Management Package (Total 1)



Item No	Item Name	Person hours*	% of Total Effort
	Iteration 1: (Due in Week 6)		
1	Introduction to project	1	2.90%
2	Project plan and summary of meeting minutes summarising the work completed so far in the project	0.5	1.45%
3	Critical Requirement Analysis (CRA)	1	2.90%
4	Package diagram and identifying who is working on each package	0.5	1.45%
5	Actor list, Actor hierarchy diagram and in-progress actor documentation— (minimum 4 actors and 1 actor documentation progress per package).	1	2.90%
6	Use case diagrams and in-progress use case documentation— (1 use case diagrams per package and 1 in-progress use case documentation per package).	1.5	4.35%
7	Activity diagrams (one activity diagram for each package) and work completed.	1.5	4.35%

8	Presentation slides and preparation	2	5.80%

Iteration 2: (Due in Week 9)

9	Executive Summary	0.5	1.45%
10	Introduction, Scope	0.5	1.45%
11	Project management documents including project plan, minutes, etc.as specified in the project template	1	2.90%
12	Critical Requirement Analysis (CRA) & Package Diagram	1	2.90%
13	Actor List and Documentation — (minimum 3 Actors per package)	1	2.90%
14	Use case documentation — (minimum 4 use case per package 16 use cases for 4 packages).	1.5	4.35%
15	Comprehensive use case documentation including use case thumbnail, use case description, stereotype and package, actors, pre-conditions, post-conditions, use case relationships, basic flow (or normal course of events), alternative flow, exceptions, user interface specification, priority, status, author and history, and reference material.	1.5	4.35%
16	Use case diagrams — (2 use case diagrams per package. 8 use case diagrams for 4 packages).	1.5	4.35%
17	Activity Diagrams — (4 activity diagram for each package 16 diagrams for 4 packages)	1.5	4.35%
18	Class documentations - (2 class documentations for each package including around 5 attributes and 5 operations for each class, 8 class documentations for 4 packages). Comprehensive class documentation including stereotype & class name, description, relationship, attributes, responsibilities, business rules and complexity	1	2.90%
20	Class diagram — (around 4 to 6 class diagrams for 4 packages including around 5 attributes and 5 operations for each class)	1.5	4.35%
21	Sequence diagrams — (2 diagrams per package, 8 diagrams for 4 packages.)	1.5	4.35%
22	Interaction overview diagrams — (1 diagram per package, 4 diagrams for 4 packages)	1	2.90%
23	State machine diagrams — (1 diagrams per package, 4 diagrams for 4 packages)	1	2.90%
24	Revised project plan and meeting minutes documentation-combining group work and quality control	2	5.80%